

Visual Search Lab

Instructions

The goal of this lab is to ask a **research question** about how humans find what they are looking for. You should identify a question about visual search that can be answered by making a small modification to the example experiment. How do humans find what they are looking for?

This should be an **experimental** study, and your research question should require comparing the **search cost** (estimated as the slope of a linear fit to reaction times) between two or more conditions. You need **at least one independent (predictor) variable** in addition to a set size variable. You may include a participant variable, such as age, gender, or other prior experience, but at least one IV must be manipulated by the researchers.

Before developing your experiment, you have some decisions to make. With your group, discuss and decide answers to the following procedure questions. For all of them, consider how different alternatives might affect *reliability*, *internal validity*, and *external validity* of your testing. (Note that as you consider later questions, you may choose to change your answers to the earlier ones.)

- What aspect of visual search are you interested in exploring? You might explore color, size, shape, complexity, semantic meaningfulness, or something else.
- What specific comparison (and thus specific research question) seems most interesting and feasible? This should be of the form “Do visual search costs differ between X and Y?”
- How will you operationalize your primary predictor variable? What are the levels of this predictor, and what stimuli will you use? How will you create them?
- Will your stimuli differ between conditions in any way other than the predictor you are trying to manipulate? How can you minimize these confounds?
- In order to estimate a search cost, your participants need to perform the same visual search task with different numbers of distractors on the screen. What set sizes will you use?

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- How many trials of each condition (one set size at one level of your primary IV) will subjects complete? We recommend that you target a total experiment duration of 10-15 minutes, including collecting informed consent and any instructions.
- Sketch out a flowchart of the order of events (“Screens”) within one trial.
- Sketch out a flowchart of the order of events (“Tasks” / “Displays”) within the entire experiment.

After you have answered these questions, move on to creating your experiment in Gorilla. Complete any necessary updates to the stimuli, displays, control spreadsheet, consent form, and experiment builder.